

Measuring and Recording the Tide in the Shark River

Tides play an important role in the marine life along our shores; understanding what influences tides and then using this knowledge to predict tide levels is important not only to monitoring marine ecology, but for navigation and boating safety as well. Our Shark River Estuary is a unique waterway a stone's throw away from Info Age. Measuring the tides offers us the opportunity to learn more about it and to connect with students interested in science, technology, and our environment.

At the InfoAge Maker Space, the CDL, we are developing a program to build and deploy tide sensors, connected by radio, to record the changing tide level at several locations of the Shark River estuary throughout the year. The sensors we are designing are to be robust, independently operating, and unobtrusive. We plan to collect this data at a receiver at Camp Evans so we can explore how the moon, sun, wind speed and direction, and rainfall influence the changing tides.

The CDL members bring expertise in electronics, radio, 3D printing, software development, and design to bear on this project. It offers an opportunity to create education programs where students can explore a wide variety of scientific and engineering topics including radio communications, electronics, marine ecology, data analysis, and machine learning.